Batterjee Medical College
For Science and Technology

The pituitary gland Disorders

Internal Medicine Department

Learning Objectives

Knowledge

- Relate pathophysiological basics and etiology of pituitary gland disorders
- Describe epidemiology, manifestations, complications and management of pituitary gland disorders

Skills

- Interpret clinical manifestations and investigations of pituitary gland disorders to formulate reasonable diagnosis.
- Apply evidence-based management plan for pituitary gland disorders
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

Main Subtopics

- Hypothalamus and pituitary gland (anatomy, blood supply, histology, hormonal regulation)
- Pituitary tumors (incidence, types, clinical picture, mass effects, local complications)
- Hyperprolactinemia and galactorrhea (causes, differential diagnosis, treatment)
- Acromegaly and gigantism (clinical picture, differential diagnosis of tall stature, investigations, treatment)

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Pituitary Gland





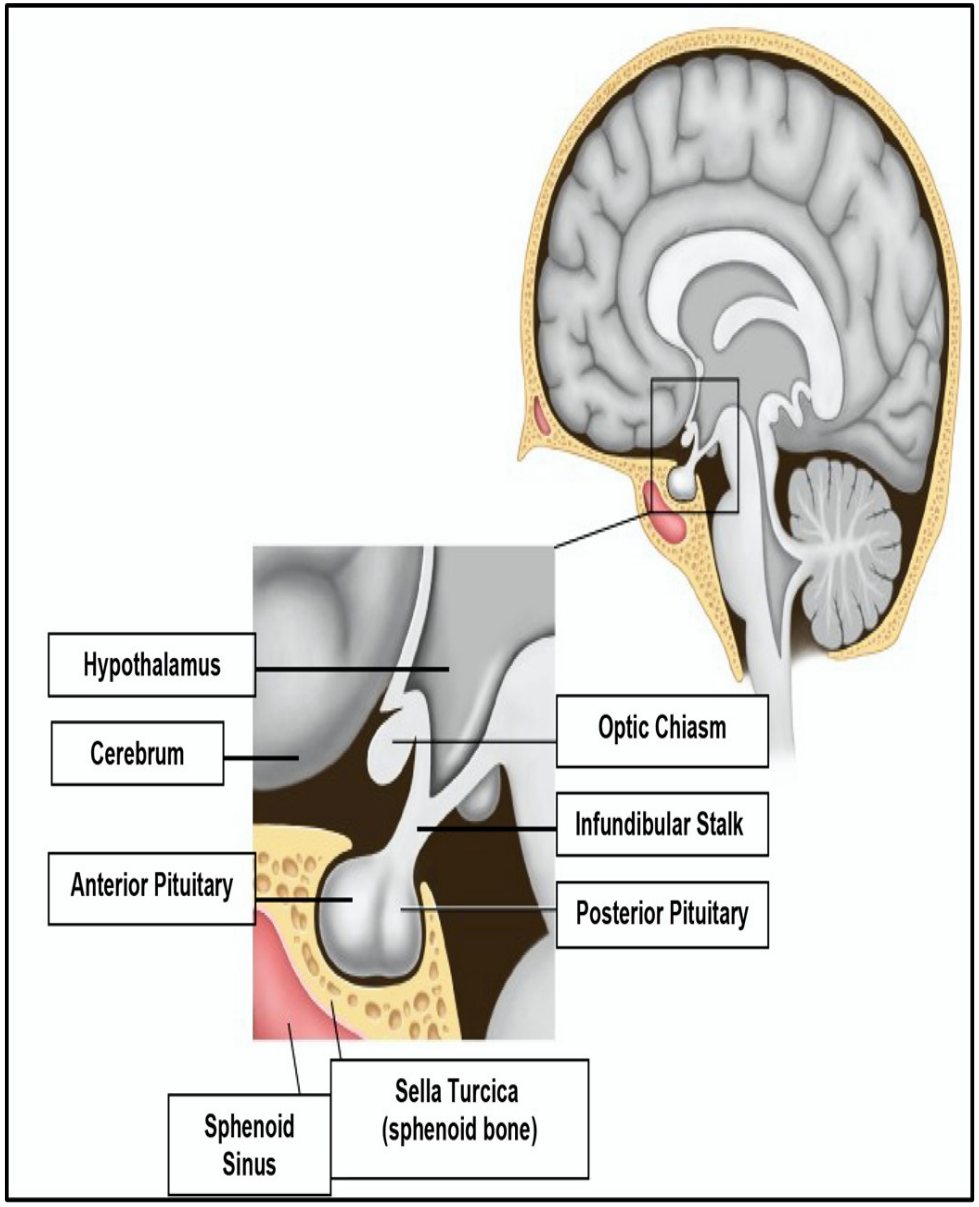
Anatomy

1) Site & relations:

- Lies within sella turcica.
- The dura mater lines the sella turcica and forms a hole at its roof (diaphragma sella) through which passes the pituitary stalk (connecting pituitary with the hypothalamus).

2) Weight:

- 1/2 - 1 gm (6 -13 mm in each dimension).



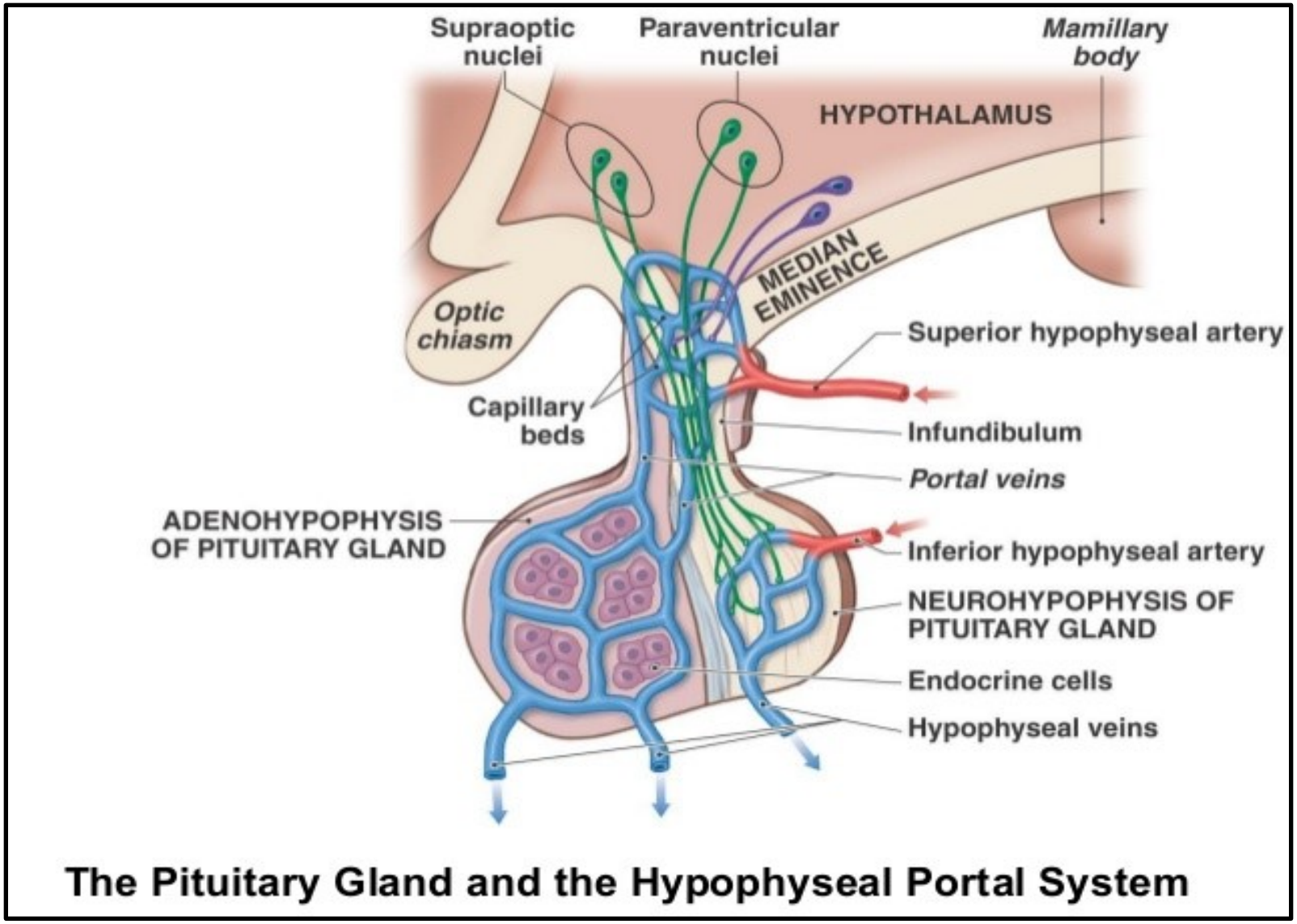
3) Blood supply:

A- Hypothalamus: superior hypophyseal artery (branch from ICA).

B- Pituitary portal circulation :

- The hypothalamus drains into the long portal veins which are the main blood supply of the anterior pituitary.
- The long portal veins form capillary plexuses in the ant. Pituitary.
- These plexuses drain into the short portal veins (which constitute the main venous drainage).

C- The posterior lobe: through inferior hypophyseal artery.



The Pituitary Gland and the Hypophyseal Portal System

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Anterior Pituitary

Histology

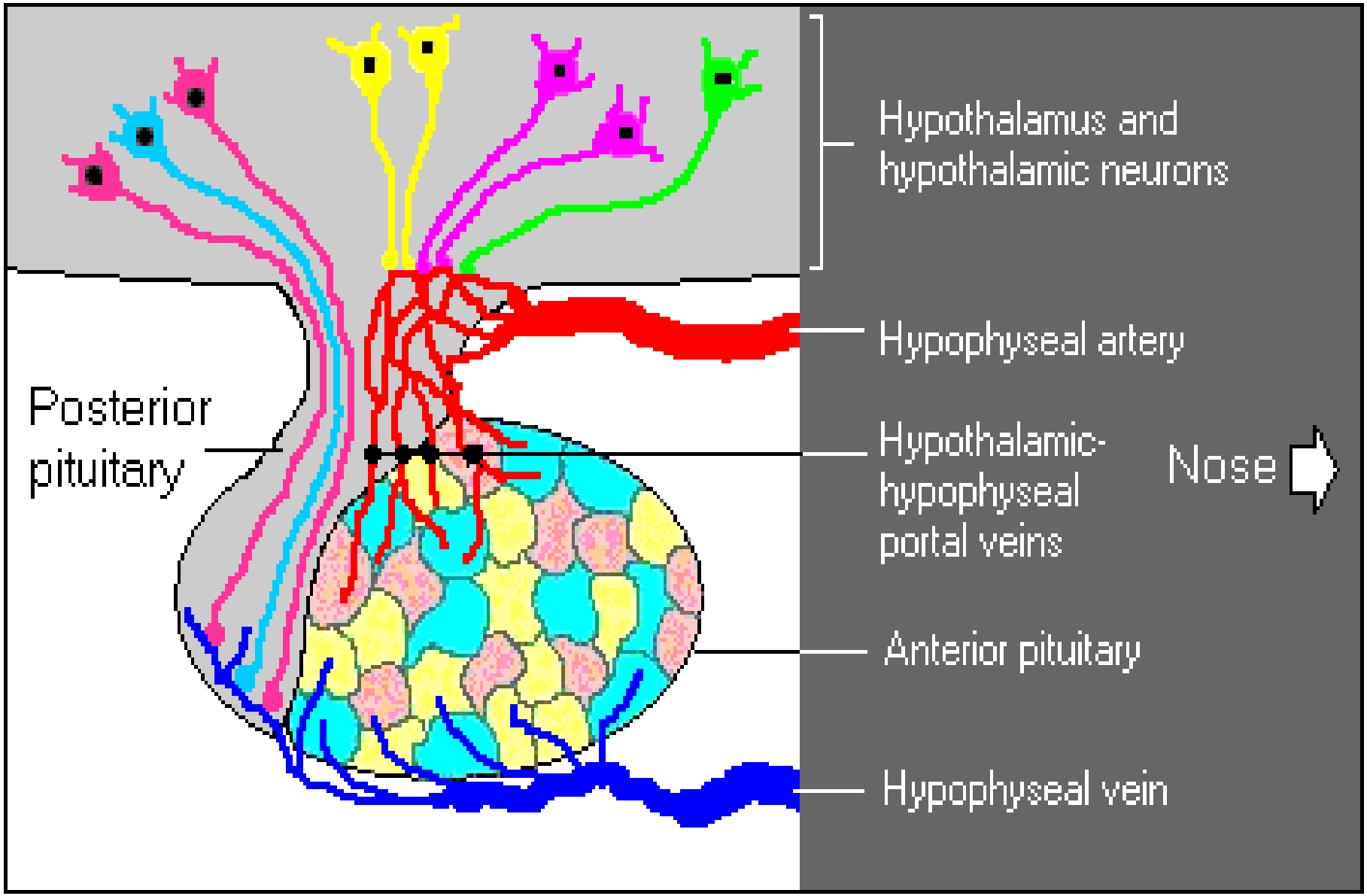
anterior lobe: (contain 3 types of cells A, B, C).

1) Acidophils: prolactin [PRL], Growth hormone [GH].

2) Basophils: TSH, ACTH, FSH → (++ maturation of grafian follicle & spermatogenesis) & LH → (ovulation & corpus luteum formation & ++ testosterone secretion).

3) Chromophobe cells: no secretions.

N.B: Posterior pituitary hormones (oxytocin & ADH) are synthesized in paraventricular & supraoptic nuclei "of the hypothalamus" respectively.


Hypothalamus and
hypothalamic neurons
Hypophyseal artery
Posterior
pituitary
Hypothalamic-
hypophyseal
portal veins
Nose
Anterior pituitary
Hypophyseal vein

Regulation : " of anterior pituitary secretion"

1.hypothalamic regulatory hormones :

- TRH (thyrotropin releasing hormone) → ++ release of TSH , PRL
- LH- RH (Gn RH) →++ release of LH, FSH " LH > FSH"
- GH-RH→++ release of GH.
- GH-RIH (somatostatin) :
  - It is also present in the D cells of islets of Langerhans.
  - ↓↓ growth hormone secretion.
  - ↓↓ secretion of insulin, glucagon, gastrin, VIP.
- CRH (corticotropin releasing hormone) : ++ release of ACTH.

***NB**: Prolactin has no specific releasing factor but dopamine is the probable PRL inhibiting factor.*

2.Feedback on pituitary & hypothalamus : "long feed-back loop"

By the hormones of target glands e.g.→ Thyroxin on TSH response to TRH.

3.Short feedback : by the pituitary hormone itself on the hypothalamus 'short feedback loop

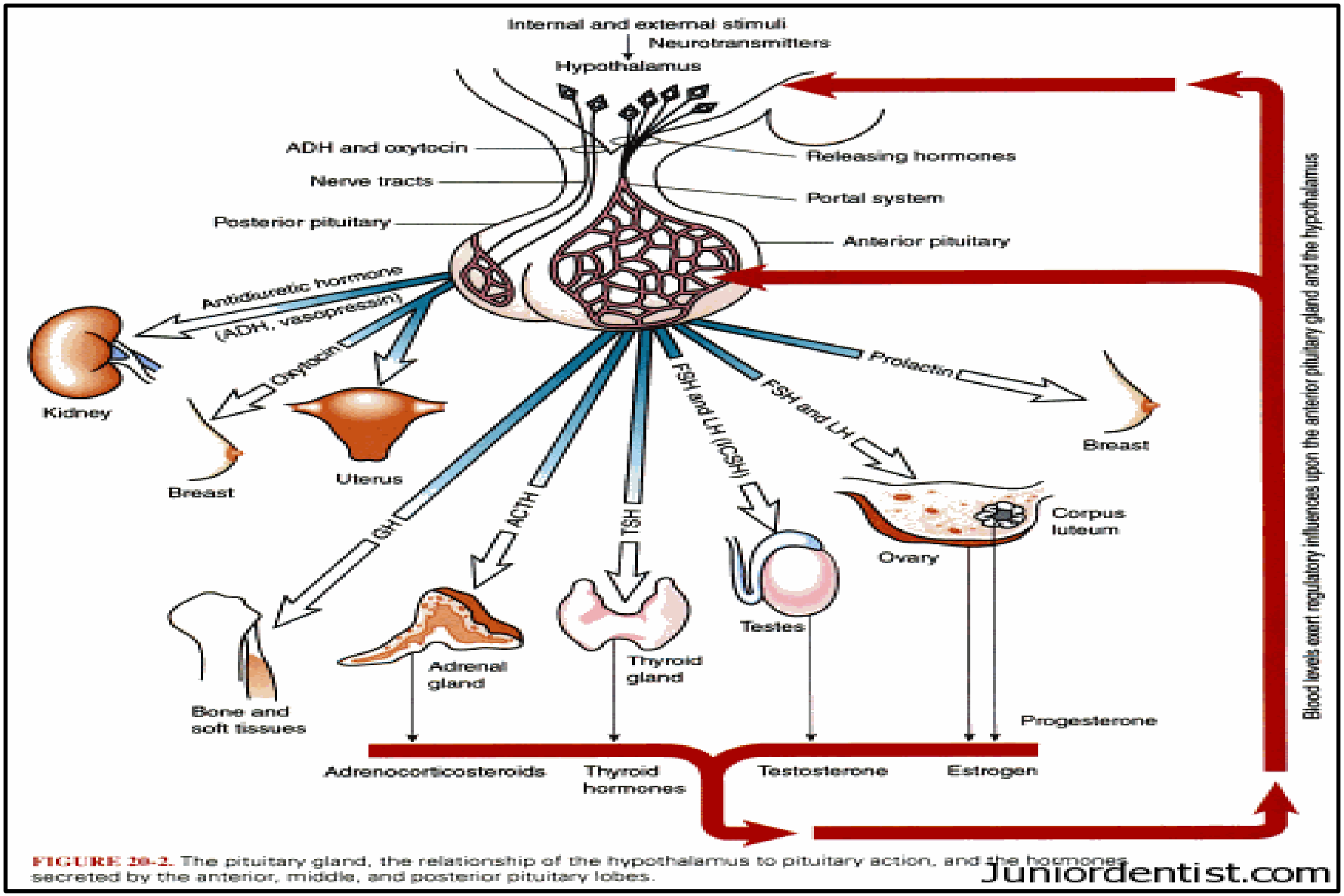


FIGURE 20-2. The pituitary gland, the relationship of the hypothalamus to pituitary action, and the hormones secreted by the anterior, middle, and posterior pituitary lobes.

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PITUITARY TUMOURS

Incidence

1.Constitute about 15 % of all intracranial tumors.

2.Types:

- PRL producing tumours→ 35 %
- Non-functioning tumours → 30 %
- GH secreting tumours → 20 %
- Mixed PRL & GH secreting tumours →10 %
- TSH , LH , FSH , ACTH secreting tumours →5 %

Clinical picture:

- Acute : Pituitary apoplexy
- Chronic :

➢ **Endocrine effects** :

- Hyperpituitarism (according to the hormone secreted).
- Hypopituitarism.
- Associations : MEN I:(pituitary, Pancreas Parathyroid).

➢ **Mass effects**

1.Headache: bitemporal

NB: When the tumour enlarges, it erodes the clenoid pr. & escapes to the cranial cavity → improval.

Rare and late, it increases in size → occipital headache & ↑ ICT

2.Superior extension

- Chiasmal syndrome : ↓ Visual acuity & bitemporal hemianopia.
- Hypothalamic syndrome :
  - o DI.
  - o Disturbance of→ thirst, temp. , appetite & sleep regulation.
  - o SIADH

3.Lateral extension

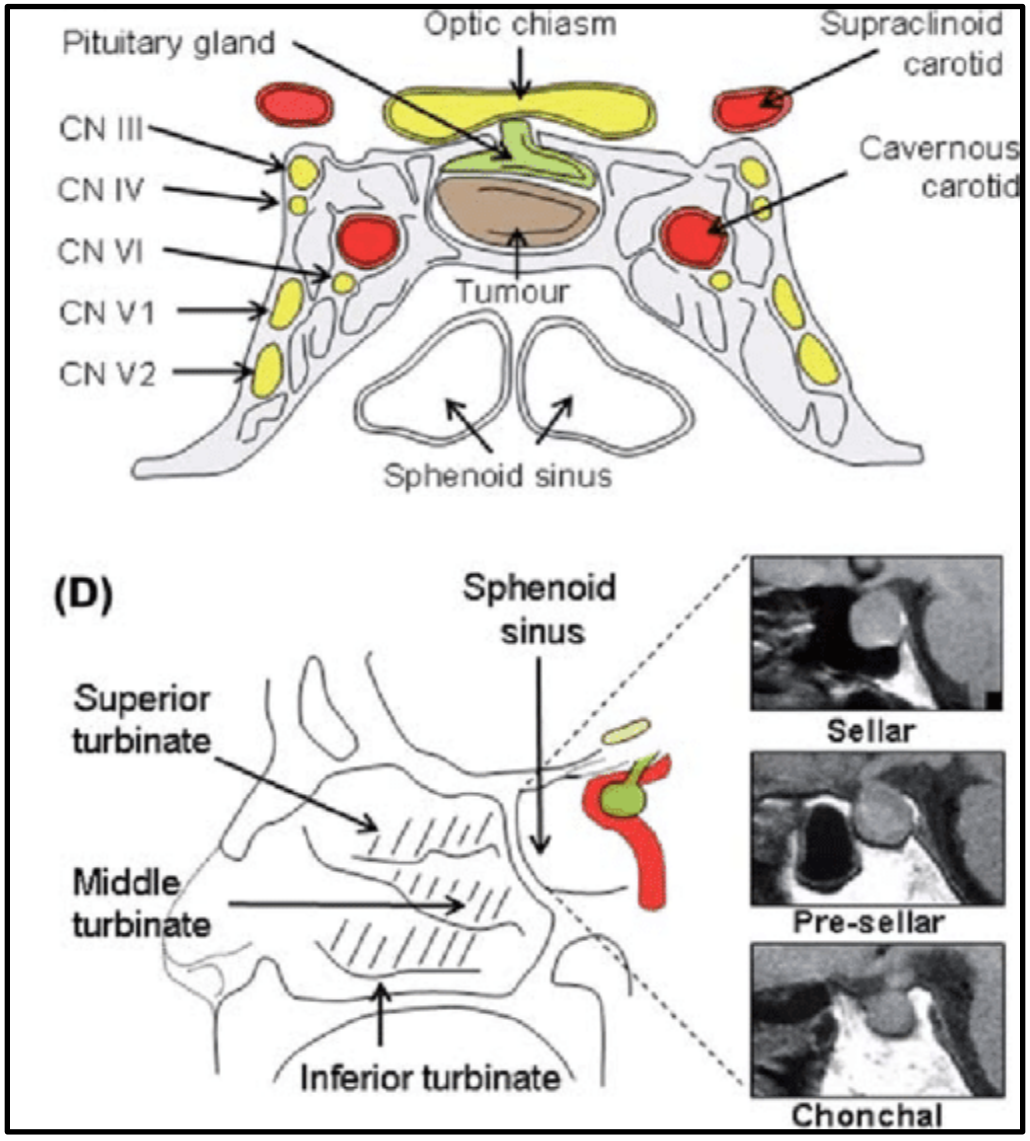
- Cranial 3, 4, 6 & ophthalmic division of trigeminal n. impairment →diplopia & facial pain.
- Temporal lobe dysfunction.

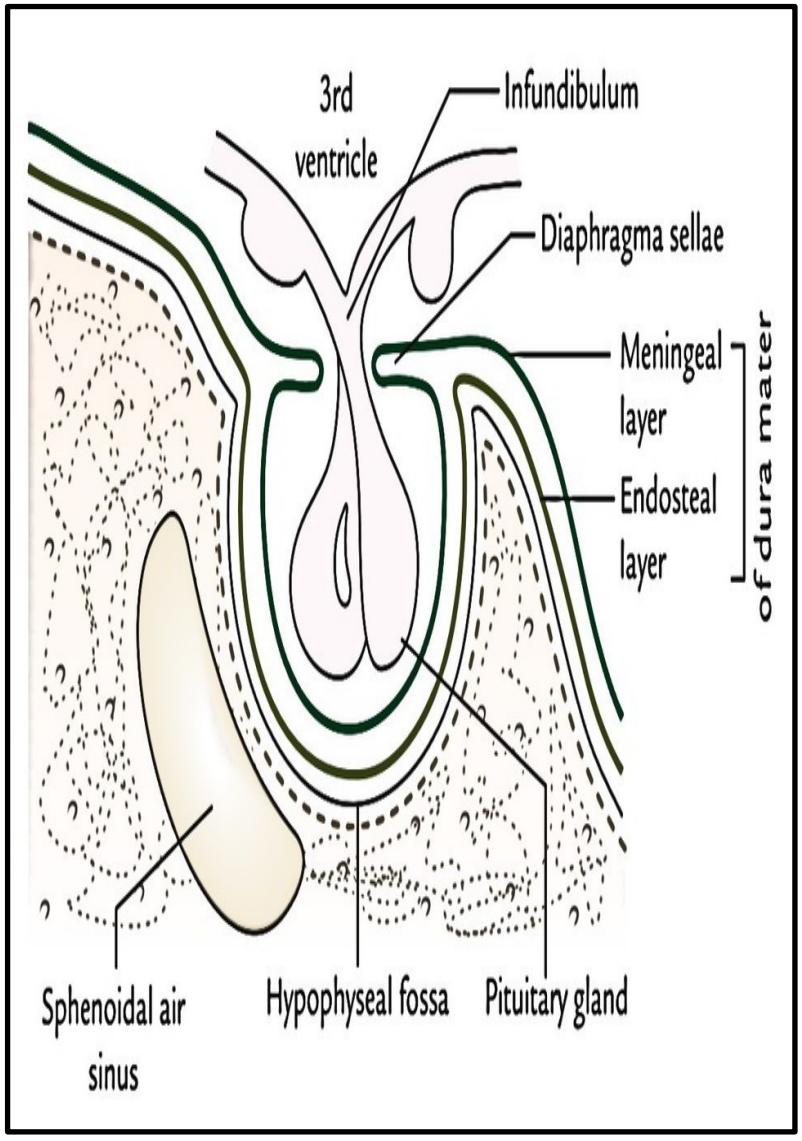
4.Inferior extension

- Nasopharyngeal mass
- C.S.F. rhinorrhoea

5.Posterior extension :

- Bilateral Δ signs.


Pituitary gland
Optic chiasm
Supraclinoid carotid
CN III
CN IV
CN VI
CN V1
CN V2
Cavernous carotid
Tumour
Sphenoid sinus
(D)
Sphenoid sinus
Superior turbinate
Middle turbinate
Inferior turbinate
Sellar
Pre-sellar
Chonchal


3rd ventricle
Infundibulum
Diaphragma sellae
Meningeal layer
Endosteal layer
of dura mater
Sphenoidal air sinus
Hypophyseal fossa
Pituitary gland

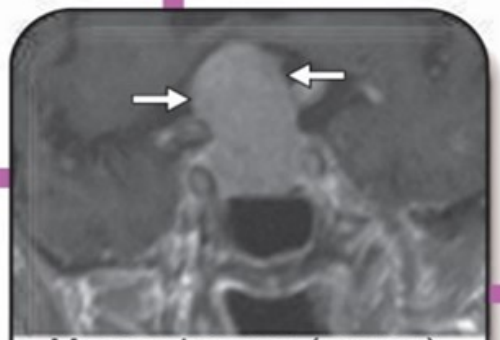
Local complications
- Headache
- Visual field defect
- Disconnection hyperprolactinaemia
- Diplopia (cavernous sinus involvement)
- Acute infarction/expansion (pituitary apoplexy)

Hormone excess

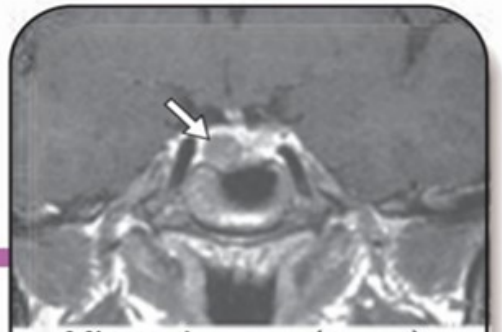
Hyperprolactinaemia
- Galactorrhoea
- Amenorrhoea
- Hypogonadism

Acromegaly
- Headache
- Sweating
- Change in shoe and ring size

Cushing's disease
- Weight gain
- Bruising
- Myopathy
- Hypertension
- Striae
- Depression



Macroadenoma (arrows) >10 mm diameter



Microadenoma (arrow) <10 mm diameter

Hypopituitarism

Growth hormone
- Lethargy

Gonadotrophins
- Lethargy
- Loss of libido
- Hair loss
- Amenorrhoea

ACTH
- Lethargy
- Postural hypotension
- Pallor
- Hair loss

TSH
- Lethargy

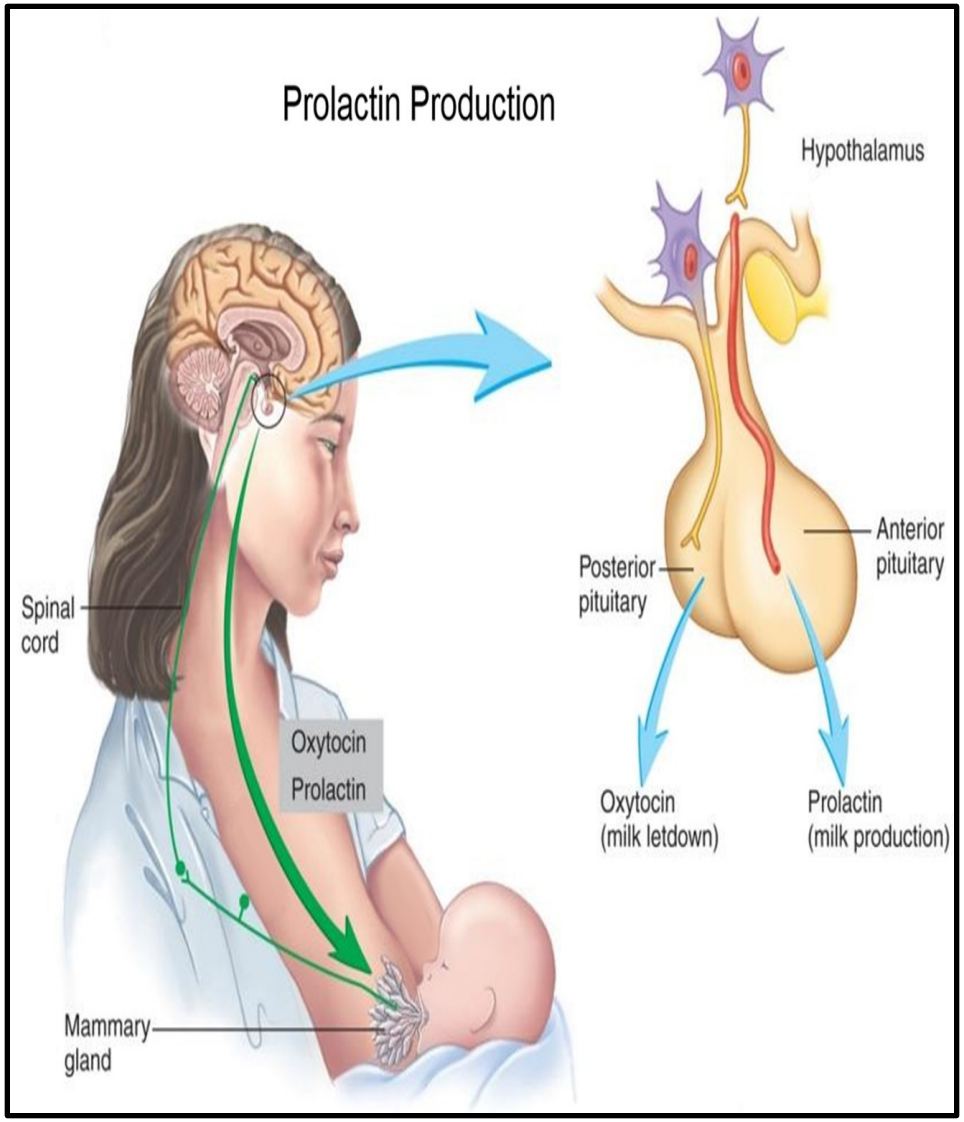
Vasopressin (ADH) (usually post-surgical)
- Thirst and polyuria

PROLACTIN HYPERSECRETION

Physiology of Prolactin:

- Actions :
  - o Breast milk production.
  - o Suppression of menstruation.

- Control: → Nocturnal peak (which is sleeping related).



DD. of causes of Hyperprolactinaemia

A. Physiological:

- pregnancy.
- 3S : suckling - stress - stimulation of the nipple.

B. Drugs

- Dopamine depleting agents e. g. methyl-dopa, reserpine.
- Dopamine receptor antagonists e.g. phenothiazines (chlorpromazine) & antipsychotic drugs
- Oestrogen opiates.

C. Pathologic causes: 3C , 3P

- CRF.
- Cirrhosis.
- Chest wall trauma.
- Pituitary (prolactinoma & tumours with pituitary stalk).
- Production from other tumours “Paramalignant”.
- Primary hypothyroidism.
-

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PROLACTINOMA

Clinical picture

This is usually a microadenoma in ♀, macroadenoma in ♂.

1. Manifestations of sellar tumour → see before
2. In females: (↑ prolactin → ↓↓ GnRH)
   - Amenorrhea and/or galactorrhea.
   - Hypoestrogenic symptoms :
     - o ↓ Libido, ↓ Vaginal secretion, Dyspareunia.
     - o Osteoporosis.
   - Hyperandrogenic manifestations :
     - o Hirsutism & acne (↑ PRL → ↑ adrenal androgens)
3. In males :
   - ↓ Libido & impotence.
   - Oligospermia & infertility.
   - Gynaecomastia ‘rare’.
4. In both sexes:→delayed puberty.

Investigations

- 1.Basal prolactin level:
  - ♀ →0-25 ng/ ml
  - ♂ → 0-20 ng/ ml

Clinical pearl

Serum prolactin levels < 200 ng/mL suggest a cause other than a prolactinoma, e.g., pharmacological or primary hypothyroidism. [2] [3]

Clinical pearl

Always review the patient's medication history for potential causes of hyperprolactinemia.

- 2.Plain X-ray, CT, MRI →for detection of pituitary tumours

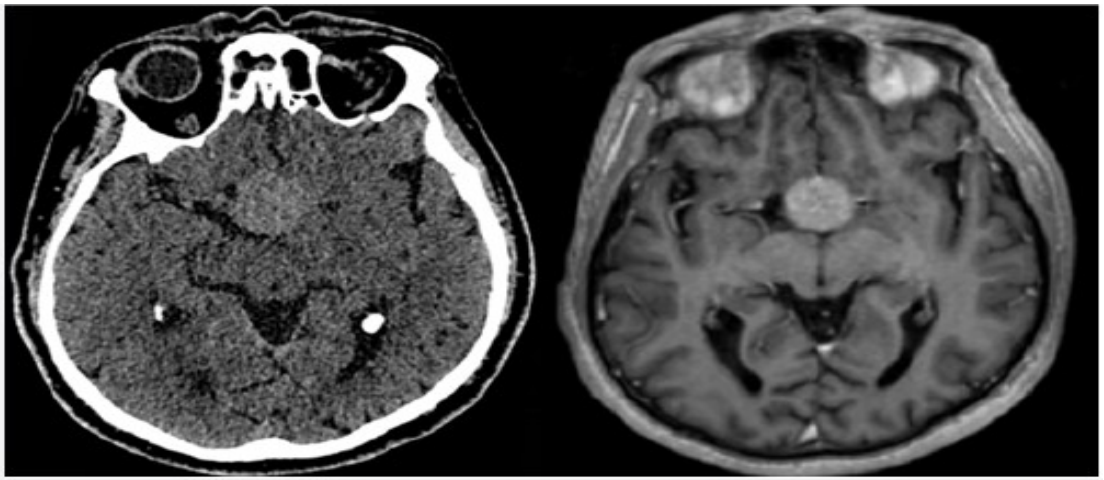
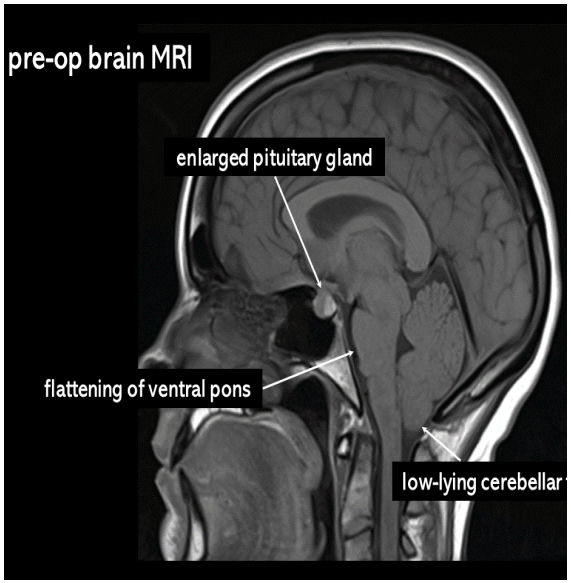


Figure 1: Treatment planning CT and MR images of a patient with pituitary adenoma.



Clinical pearl

In patients with endocrine dysfunction, order hormone assays before imaging to prevent overdiagnosis of pituitary incidentalomas. [10]

Treatment:

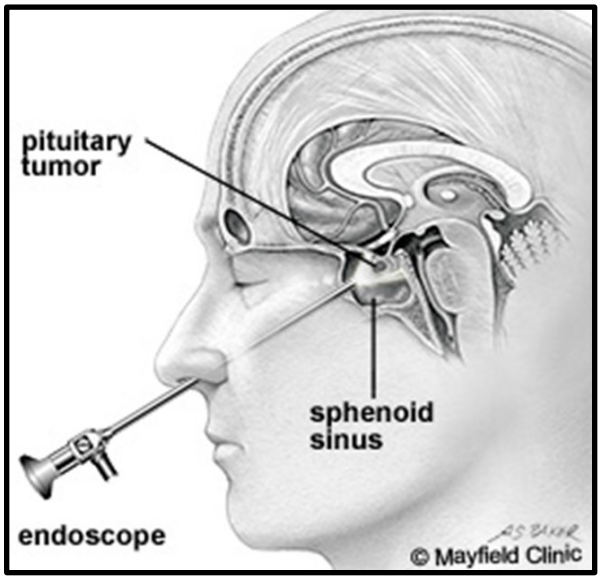
1) Surgery:

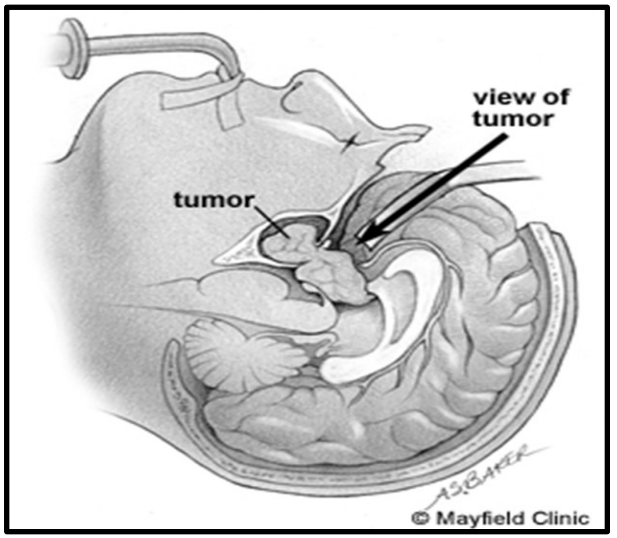
A. Trans-sphenoidal: for micro-adenoma (< 10 mm).

- Complications:

- Hypopituitarism
- C.S.F rhinorrhea.
- Meningitis.
- Optic n. damage

B. Trans-frontal: for large tumours (> 10 mm).





2) Radiotherapy: 4500 rads.

3) Medical ttt → Bromocriptine 'dopaminergic agonist'.

- Aim of ttt :
  - o ↓ hyperprolaotinaemia due to any cause.
  - o Restores menstruation, fertility & stops galactorrhea.
  - o Anti-tumour effect →↓size of the tumour.
- Dose→2.5 mg t.d.s with meals (given for I - 2 years).
- Side effects → nausea, vomiting & postural hypotension.

N.B: Recently, cabergolin is more effective but expensive.

Initial Treatment of pituitary adenomas:

Tumor type	First-line treatment
Prolactinomas (symptomatic or macroadenomas)	Pharmacotherapy
Secretory adenomas (except prolactinomas)	Surgery
Symptomatic nonsecretory adenomas	
Asymptomatic microprolactinomas	Observation
Asymptomatic nonsecretory adenomas	

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Growth Hormone Hypersecretion

Physiology of GH

↑ Blood glucose

- ↑ Protein synthesis.
- ↑ Free fatty acids ' FFA'

It acts through induction of somatomedins (which are mainly synthesized by the liver).

Control of GH

Stimulation (via GHRH)	Suppression (via somatostatin)
• Hypoglycemia. • Stress, exercise. • Estrogen, clonidine, L-dopa.	• Hyperglycemia. • Obesity.

GH hypersecretion: is due to acidophil pituitary adenoma:

either:

Acromegaly	or	Gigantism
(After fusion of epiphyses)		(Before fusion of epiphyses)

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Gigantism

- There is massive growth of the skeleton (height up to 2,5 m).
- Soft tissues also enlarge.
- Early, the patient is strong but later on, suffers from weakness, hypogonadism, myopathy, peripheral neuropathy (decline stage).

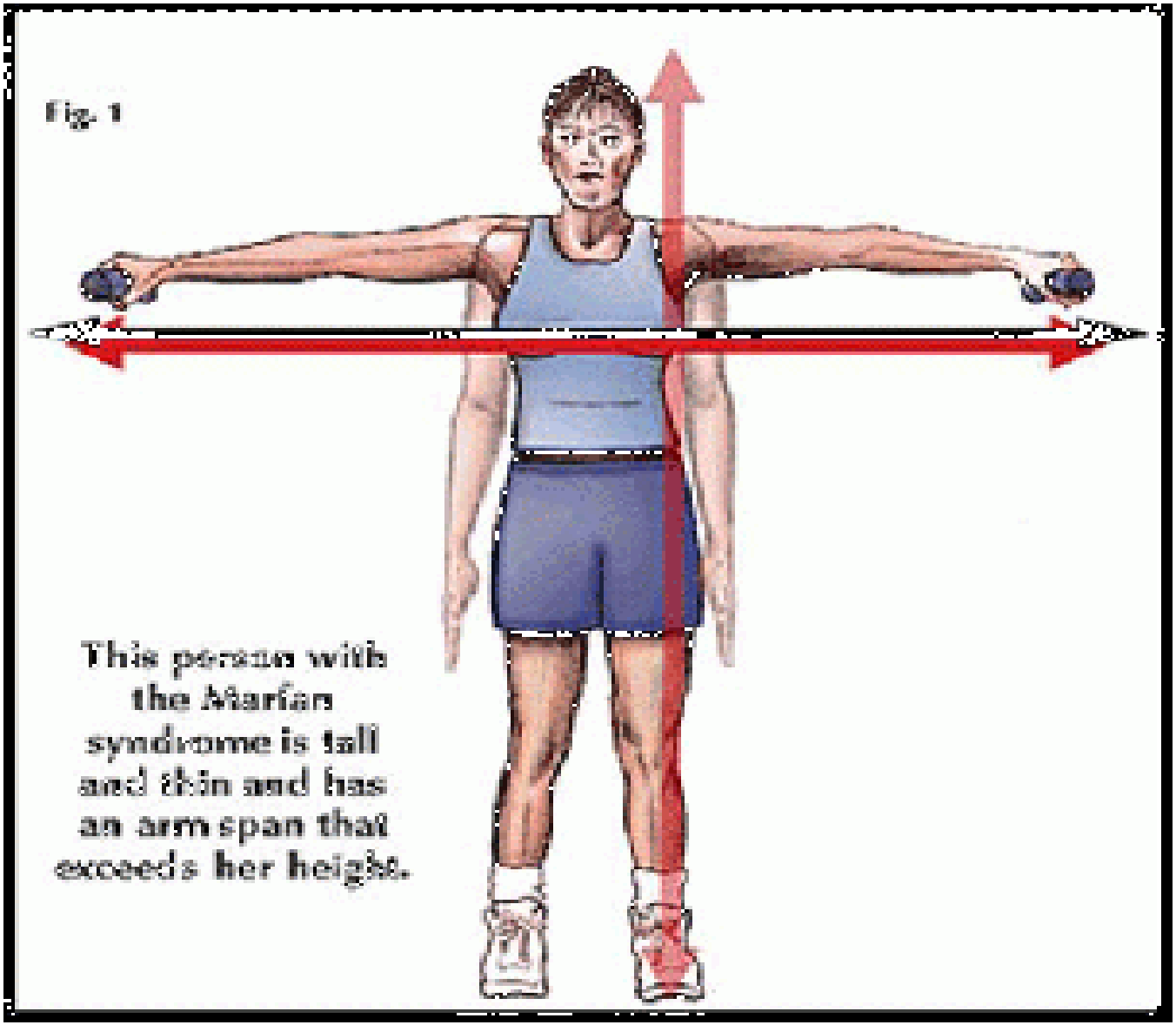





CONGRESS OF FREAKS with RINGLING BROTHERS and BARNUM & BAILEY COMBINED CIRCUS
1931
CENTURY

Differential diagnosis (other causes of tall stature)

- Constitutional / genetic (most common) → proportionate with normal GH level.
- Cerebral gigantism (Soto's syndrome)→ Early puberty .
- Eunuchoidism(disproportionate + features of hypogonadism).
- Hyperthyroidism.
- Homocystinuria (like marfan+ mental retardation).
- Marfan's syndrome.
- Klinefelter syndrome → Azospermia, gynaecomastia, +ve buccal smear.


Fig. 1
This person with the Marfan syndrome is tall and thin and has an arm span that exceeds her height.

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Acromegaly

- It is so called because there is enlargement of the acral parts (hands, nose, feet).

- Slowly progressive starting between the 2nd -4th decades (manifestations appear 10 - 20 years later).

Clinical picture:

1) Facial features:

o Enlarged lips & nose (and soft tissue of the face).

o Enlargement of mandible → prognathism & ↑ spacing of lower teeth.

o Enlargement of tongue→(macroglossia).

o Enlargement of frontal sinuses → prominent forehead.

o Voice→ cavernous & husky.

2) Soft tissue overgrowth :

o Leading to ↑ ring, glove, shoe and hat size.

o Hands & feet are grossly enlarged(spade- like hands).

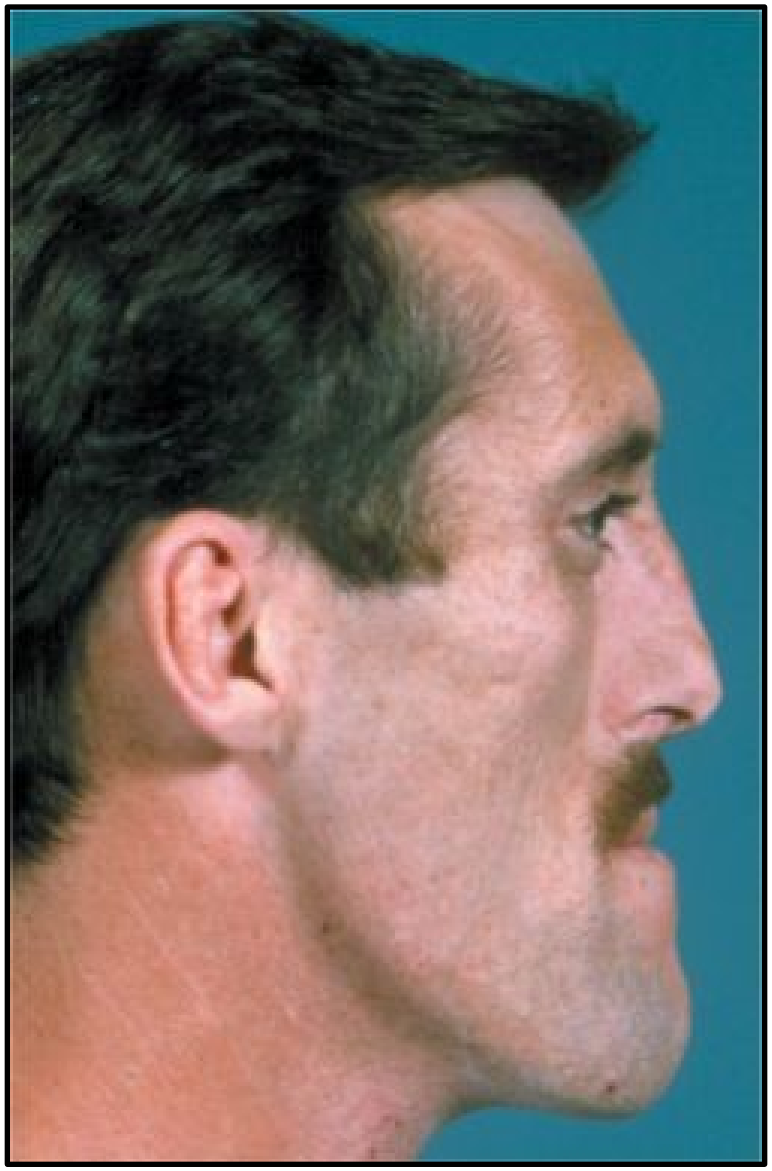






Maurice Tillet (Swedish Angel) checks head sizes with Marj.










B



3) Neurological:

- o Entrapment neuropathy (carpal tunnel syndrome).
- o Peripheral neuropathy.
- o Neuropathies 2ry to DM.
- o Proximal myopathy.
- o Manifestations due to GH producing tumour in the skull

4) Visceromegaly: eg. Liver, spleen, kidney, … etc

5) Joint changes: osteoarthritis (especially of the spine, hip, knee) “late”.

6) CVS:

- o Hypertension.
- o Cardiomegaly.
- o Raynaud's phenomenon.
- o Conduction defects & arrhythmia.
- o Sleep apnea.

7) Skin changes :

- o Hypertrichosis " ↑ growth of coarse body hair "
- o Increased pigmentation, acanthosis nigricans.
- o Warm, moist, thickened (+ acne & oiliness)

8) Metabolic effects:

- ↑ BMR → heat intolerance & ↑ sweating (hyperhidrosis).
- Glucose intolerance in 50 % of cases & frank diabetes in 25 %.
- Hyperphosphataemia.
- Hypercalcaemia (in 16 % of cases) due to ↑ l,25 (oH)2 D3.

9) Associated endocrine features :

- Associated pituitary tumours :
  - PRL→hyperprolactinaemia.
  - ACTH → Cushing's disease.
  - TSH →2ry thyrotoxicosis.
- Associated MEN type I.
- DI (due to suprasellar extension of the tumour).

Investigation

- **Radiological**:

1) Skull X—ray shows:

o Thickening of the cortex.

o Pneumatization of the frontal sinus.

o Prominent external occipital protuberance.

o Frognathism & big mastoid process.

o Features of intrasellar tumour: Double floor (early) then

ballooning & erosion of clinoid processes.



2) X-ray of the hands :

Tufting of the terminal phalanges

(brush border) ‘due to periosteal thickening

‘Mushroom appearance→ pseudoclubbing



3) X-ray on the heel:

Thickening of the soft tissue of the heel pad > 22 mm.

4) Joints (X-ray) →osteoarthritic changes.

5) CT. & MRI →for detection of pituitary tumours.

NB.: X-ray chest can be done to exclude bronchial carcinoid as a cause of acromegaly (paramalignant syndrome).

- **Visual field & ophthalmologic exam :** bitemporal hemianopia, papilloedema, optic atrophy, glaucoma ' in severe cases

> ⚠ Caution
>
> Visual defects may be present even in asymptomatic patients with a pituitary adenoma; consider early referral for visual field testing even in the absence of visual symptoms. 9

- **Endocrinal**

> ⚠ Caution
>
> Random GH measurements are not recommended for the diagnostic evaluation of acromegaly. 2

1) GH.

o ↑ Basal level " > 6 ng /ml ". GH is difficult to measure as of its pulsatile secretions

o IGF-1 levels are increased. (Best initial)

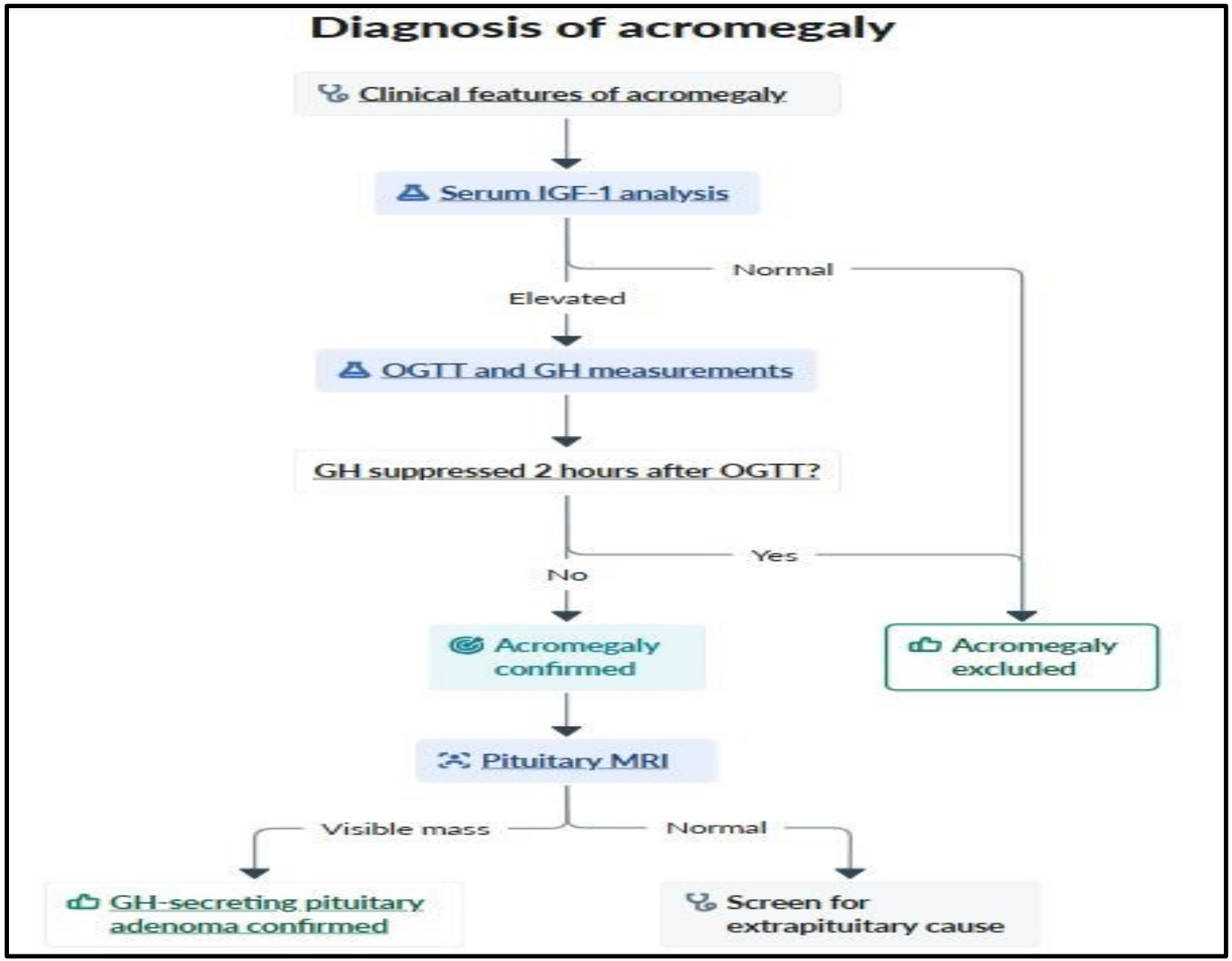
o Failure of GH. suppression in response to oral glucose load (to < 2 ng/ml). (Most accurate)

2) GH-RH → ↑ in ectopic secretion from bronchial carcinoid.

- **Investigation for metabolic effects**

o ↑ BMR o Hyperglycaemia. o Hypercalcaemia.

- **Echocardiography and colonoscopy:** should be performed to evaluate for cardiomegaly and colon polyps.


Diagnosis of acromegaly
Clinical features of acromegaly
Serum IGF-1 analysis
Normal
Elevated
OGTT and GH measurements
GH suppressed 2 hours after OGTT?
Yes
No
Acromegaly confirmed
Acromegaly excluded
Pituitary MRI
Visible mass
Normal
GH-secreting pituitary adenoma confirmed
Screen for extrapituitary cause

Treatment

1) Surgical: as prolactinoma.

2) Radiotherapy:

- proton beam, heavy particles & supravoltage radiation.

- may require several years for a full therapeutic effects.

3) Medical tt:

A. Dopamine agonist e.g. bromocriptine.

- indicated when surgery is contraindicated especially for cases.
- needing rapid relief of symptoms.
- dose→ 15-30 mg/day.
- GH ↓ only in 10- 40 % of cases.
- it is used as adjunctive therapy following surgery or radiotherapy.

> **Clinical pearl**
>
> SRLs and dopamine agonists can reduce both GH secretion and tumor size, while GH receptor antagonists block IGF-1 production and do not affect tumor size. 2

B. Somatostatin:

- Not specific for GH.
- Has to be given by infusion.

C. Octreotide (analogue of somatostatin) :

- Has longer duration of action & is specific.
- Can be given s.c. / 12 hrs

Case scenario :

A 32-year-old woman comes to the physician because of a 3-month history of irregular menses, milky discharge from her nipples, fatigue, and weight gain. Menses occur at irregular 25–40-day intervals and last 1–2 days with minimal flow. 5 months ago, she was started on clozapine for treatment of schizophrenia. She has hypothyroidism but has not been taking levothyroxine over the past 6 months. Visual field examination show no abnormalities. Her serum thyroid-stimulating hormone is 17.0 µU/mL and serum prolactin is 85 ng/mL.

Which of the following is the most likely explanation for the nipple discharge in this patient?

A Hypothyroidism

B Prolactinoma

C Thyrotropic pituitary adenoma

D Ectopic prolactin production

E Adverse effect of medication

Case scenario :

A 40-year-old man comes to the physician for the evaluation of episodic headaches for 5 months. The headaches involve both temples and are 4/10 in intensity. The patient has been taking acetaminophen, but the headaches did not subside. He has also had visual disturbances, including double vision. He has no nausea, temperature intolerance, or weight changes. The patient does not smoke. He drinks 2–3 beers on weekends. He appears pale. His temperature is 37°C (98.6°F), pulse is 75/min, and blood pressure 125/80 mm Hg. Ophthalmologic examination shows impaired peripheral vision bilaterally. An MRI scan of the head with contrast shows a 16 × 11 × 9 mm intrasellar mass.

Further evaluation is most likely to show which of the following findings?

A. Galactorrhoea

B. Coarse facial features

C. Diffuse goitre

D. Erectile dysfunction

E. Abdominal striae

Summary



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Thank You